P(1+2)

Definite Integration (3)

Q. Let $f(\theta) = 1 + (1+\sin\theta)\cos\theta + (1+\sin\theta+\sin^2\theta)\cos^2\theta + \cdots + \infty$ terms $\theta \in (0, \frac{\pi}{2})$

Q1. If value of $f(\frac{\pi}{4}) = m + n\sqrt{2}$ where $m, n \in R$, then the value of $m+n$ is greater than

(A) 3 (B) 4 (C) 5 (D) 6

Q2. The value of definite Integral $\int_0^{\pi/2} (1-\cos\theta)\cos\theta\, f(\theta)\, d\theta$ lies in the interval

(A) (0,1) (B) $(1,\sqrt{3})$ (C) $(\frac{3}{2}, 2)$ (D) (2,4)

P(3-5): Let $y = g(x)$ be a function, of the graph of broken line connected by points (-2,0) and (2,4) in the x-y plane.

Also $f(x) = \int_{-2}^{x} |g(t) - t|\, dt$, $x \in [-2,2]$ and $h(x)$ is the inverse of $f(x)$.

Q3. The value of $f'(0)$ is

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) does not exist

Q4. The value of $h(\frac{31}{4}) + h'(\frac{31}{4})$ is

(A) $\frac{17}{5}$ (B) $\frac{7}{5}$ (C) $\frac{9}{5}$ (D) $\frac{13}{5}$

Q5. The sum of roots of $f(x) = 2x^2 + 5$ is

(A) 3 (B) $\frac{1}{3}$ (C) $\frac{4}{7}$ (D) $\frac{3}{4}$

P(3-5) Let f be a real-valued continuous function on R (the set of all real numbers) and satisfying

$$f(x) = x^2 + \int_{-1}^{1} (x+y) f(y)\, dy$$

where x and y are independent variables

Q3. If $f(x) > k$ for every real x, then the largest integral value of k is

(A) -4 (B) -3 (C) -2 (D) 1

Q4. If $g(t) = f(-t) - f(t)$ for $t \in R$ then the value of $\int_0^{\pi/2} g(t) \sin t \, dt$ is equal to

(A) 0 (B) 2 (C) 3 (D) 4

Q5. If the range of function $f(x) = 3f(x) + b + 4$ for every real x is $[0, \infty)$ then b equals

(A) -1 (B) 0 (C) 3 (D) 6.

P(6-8): Let $f: R \to R$ be a differentiable function satisfying $f(x) = x^2 + 3\int_0^{x^{1/3}} e^{-t^3} t^2 f(x-t^3)\,dt$.

Q6. The area bounded by $f(x) = 0$ and $y = x^2 + 3x$ is

(A) $\frac{27}{14}$ (B) $\frac{27}{8}$ (C) $\frac{27}{4}$ (D) $\frac{27}{2}$.

Q7. The value of determinant

$$\begin{vmatrix} f(0) & (f(3)-9) & (f(5)-25) \\ (f(-3)-9) & f(0) & (f(7)-49) \\ (f(-5)-25) & (f(-7)-49) & f(0) \end{vmatrix}$$ is

(A) 0 (B) -1 (C) 2 (D) -3.

Q8. Which of the following statement is correct?

(A) $f(x)$ is an odd function (B) the value of $\lim_{x\to 0} \frac{\int_0^x (f(t) - t^2)\,dt}{x^4}$ equals $\frac{1}{24}$

(C) $f(3)$ equals 9 (D) $f'(4)$ equals 24.

P(9-11): Consider $f(x) = \int_0^x \left(\frac{f(t)}{1+e^{t-x}} + \frac{f(x-t)}{1+e^t} e^t \right) dt$

Q9. The number of solution of the equation $f(x) + x^2 = 0$ is

(A) 0 (B) 1 (C) 2 (D) 3.

Q10. If area enclosed by the curve $y = f(x)$ and x-axis from $x = x_1$ to $x = 0$ where x_1 is abscissa of point of inflexion on the graph of $y = f(x)$ is $1 - \frac{b}{e^c}$ where $b, c \in N$ then $(b + c)$ is equal to

(A) 3 (B) 4 (C) 5 (D) 7.

Q11. The value of $\lim_{x\to 0^+} (f(x))^{(f(x))^x}$ is equal to

(A) 0 (B) 1 (C) e (D) non-existent.

P(12-13): Let f be a cubic polynomial function such that $f(x) + f(-x) = 0 \ \forall\ x \in R$ and $\lim_{h\to \frac{1}{\sqrt{3}}} \frac{f'(h) - 2}{h - \frac{1}{\sqrt{3}}} = 4\sqrt{3}$.

Q12. If $\phi(x) = \int_1^x e^t f(x+1-t)\,dt + \frac{5}{4}$ then $\phi'(1)$ equals

(A) $\frac{2}{e}$ (B) $2e$ (C) $\frac{1}{e}$ (D) e

Q13. If $gof(x) = x \ \forall\ x \in R$ and $g(2) = a$, $g(16) = b$ then the value of $\int_a^b f(x)\,dx + \int_2^{16} g(x)\,dx$ equals

(A) 14 (B) 28 (C) 30 (D) 32.

P(14-15): Let $I_1 = \int_0^x \frac{t^2\,dt}{(x - \sin x)\sqrt{\alpha^2 + 2\alpha t}}$ (where α is non-zero constant) and $I_2 = \int_0^{x^2} \frac{e^t - 1 - t}{x^6}\,dt$.

Q14. If $\lim_{x\to 0} I_1 = 1$ then sum of values of α is

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

Q15. If $\alpha = 1$ the value of $\lim_{x\to 0} (I_1 + 6I_2)$ is equal to

(A) 2 (B) 3 (C) -1 (D) 0.

Q16. Let $A = \int_0^{e^2} \frac{\ln x}{\sqrt{x}}\,dx$. Identify which of the following statement(s) is/(are) correct?

(A) $A < 2\left(e - \frac{1}{e}\right)$ (B) $A < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$

(C) $A = \int_0^{e^2} + \sqrt{et}\,dt$ (D) $3 < A < 5$.

Q17. Given $f(x) - 2\int_0^{\pi/4} \frac{\sin^2 x}{\cos 5x} \cos t \cdot f(t)\,dt = \frac{\sin^2 x}{\cos 5x}$, then which of the following is/(are) correct?

(A) $\lim_{x\to \frac{\pi}{3}} f(x) = 72$ (B) $f(x)$ is periodic with period π.

(C) The equation of normal to the graph of $f(x)$ at point M whose abscissa is π is given by $x - \pi = 0$ (D) Number of solutions of the equation $f(x) = 0$ in $(0, 3)$

Q18. Let $I_1 = \int_0^{\pi/4} x^{2008} (\tan x)^{2008} dx$, $I_2 = \int_0^{\pi/4} x^{2009} (\tan x)^{2009} dx$

$I_3 = \int_0^{\pi/4} x^{2010} (\tan x)^{2010} x\, dx$ then which one of the following inequalities hold good?

(A) $I_2 < I_3 < I_1$ (B) $I_1 < I_2 < I_3$

(C) $I_3 < I_1 < I_2$ (D) $I_3 < I_2 < I_1$.

Q19. Which of the following statement(s) is (are) correct?

(A) If $f: [0,1] \to R$ is continuous and $\int_0^x f(x)dx = \int_x^1 f(x)dx \ \forall\ x \in [0,1]$ then $f(x) = 0$ for all $x \in [0,1]$.

(B) Let f be a function defined on $[a,b]$ such that $f(x) \geq 0 \ \forall\ x \in [a,b]$ and $\int_a^b f(x)dx = 0$ then $f(x) = 0 \ \forall\ x \in [a,b]$.

(C) If $f(x)$ is continuous on R then $\int_0^{\pi/2} f(\cos x)dx = \int_0^{\pi/2} f(\sin x)dx = \int_0^{\pi/2} f(\sin 2x)dx$.

(D) For $a > 0$, if f is continuous in $[-a,a]$ then $\int_{-a}^{a} f(x^2)\,dx = 2\int_0^a f(x^2)dx$.

Q20. Which of the following statement(s) is/are true?

(A) If function $y = f(x)$ is continuous at $x = c$ such that $f(c) \neq 0$ then $f(x)f(c) > 0 \ \forall\ x \in (c-h, c+h)$ where h is sufficiently small positive quantity.

(B) $\lim_{n\to\infty} \frac{1}{n} \ln\left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right) = 1 + 2\ln 2$

(C) Let f be a continuous and non-negative function defined on $[a,b]$. If $\int_a^b f(x)dx = 0$ then $f(x) = 0 \ \forall\ x \in [a,b]$.

(D) Let f be a continuous function defined on $[a,b]$ such that $\int_a^b f(x)dx = 0$, then there exists atleast one $c \in (a,b)$ for which $f(c) = 0$.

Q21. If $\int_0^{\pi/4} \frac{\ln(\cot x)\cdot(\sin 2x)^{2008}}{[(\sin x)^{2009}+(\cos x)^{2009}]^2}dx = \frac{d^u \ln d}{v^2}$, then $d+u+v$ is

Q22. If $\int_{-1}^{1} x^2\, d(\ln x) = \frac{e^k-1}{2e^l}$ then evaluate $k+l$.

Q23. If $f(x)$ is an even differentiable function, then find the integral $\int_{-2}^{2}(x^3 f(x)+x f''(x)+2)\,dx$.

Q24. If $f(\pi)=2$ and $\int_0^{\pi}[f(x)+f''(x)]\sin x\,dx$ has the value equal to 5, then find $f(0)$.

Q25. If $I = 636\int_0^{\pi/4}(\sqrt{\sin x}+\sqrt{\cos x})^{-4}\,dx$ then find I

Q26. prove that, if $y_1>0$, $y_1 \le y \le y_2$ and $\phi(y) = \int_0^{\infty}\frac{\sin xy}{x(a^2+x^2)}dx$ then $\phi''(y)-a^2\phi(y) = -\frac{1}{2}\pi$ show further that when $y>0$ and $a>0$ $\phi(y) = \frac{1}{2}\pi(1-e^{-ay})/a^2$.

Q27. prove that $\left[\int_0^1 f(x)\,dx\right]^2 \ge \int_0^1 [f(x)]^3\,dx$

Q28 Let $p(x)$ be a polynomial of degree at most 3

(A) show that there is a number c such that $\int_{-1/2}^{1/2} p(x)\,dx = \frac{1}{3}[p(-c)+p(0)+p(c)]$.

Q29. i) $\int_1^{\infty}\frac{\tan^{-1}(x-1)}{\sqrt[3]{(x-1)^4}}dx$ ii) $\int_{-1}^{1}\ln\frac{1+x}{1-x}\,\frac{x^3\,dx}{\sqrt{1-x^2}}$

iii) $\int_1^{\infty}\frac{x^2-2}{x^3\sqrt{x^2-1}}dx$.

Q30. If $\phi(x) = 3\int_0^x \frac{(1+\sec t)\ln\sec t\,dt}{\ln\sec x\,[x+\ln(\sec x+\tan x)]}$ then

i) $\phi(x)$ is even ii) $\phi(x)\to\frac{3}{2}$ when $x\to\frac{\pi}{2}$ through values less than $\frac{\pi}{2}$.

Q31. When $f(x)$ is bounded and is strictly decreasing in $(0,1)$ prove that $\int_0^{1-\frac{1}{n}} f(x)\,dx \geq n^{-1}\{f(\frac{1}{n}) + f(\frac{2}{n}) + \cdots + f(\frac{n-1}{n})\} \geq \int_{1/n}^{1} f(x)\,dx$.

Q32. Let n be a fixed integer. Let $f: R \to R$ be given by $g(x) = \begin{cases} x & \text{if } x \leq 0 \\ 2^n & \text{if } 2^n - 2^{n-2} < x \leq 2^{n+1} - 2^{n-1} \end{cases}$

Prove that $\int_0^{2^n} f(x)\,dx = \int_0^{2^n} x\,dx = 2^{2n} - 1$

Q33 (A) Show that $0 < \int_0^1 \frac{t^4(1-t)^4}{1+t^2}\,dt$ and that $\int_0^1 \frac{t^4(1-t)^4}{1+t^2}\,dt = \frac{22}{7} - \pi$.

(B) Evaluate $\int_0^1 t^4(1-t)^4\,dt$ then apply the results of (a) to conclude that

$\frac{22}{7} - \frac{1}{630} < \pi < \frac{22}{7} - \frac{1}{1260}$.

Q34. Use the identity $1+x^6 = (1+x^2)(1-x^2+x^4)$ to prove that $a > 0$, we have

$\frac{1}{1+a^6}\left(a - \frac{a^3}{3} + \frac{a^5}{5}\right) \leq \int_0^a \frac{dx}{1+x^2} \leq a - \frac{a^3}{3} + \frac{a^5}{5}$.

Q35. A function F is defined by the indefinite integral $F(x) = \int_1^x \frac{e^t}{t}\,dt$, $x > 0$.

(A) For what values of x is it true that $\ln x \leq F(x)$?

(B) Prove that $\int_1^x \frac{e^t}{t+a}\,dt = e^{-a}[F(x+a) - F(1+a)]$

(C) In a similar way, express the following integrals in terms of F:

$\int_1^x \frac{e^{at}}{t}\,dt$, $\int_1^x \frac{e^t}{t^2}\,dt$, $\int_1^x e^{1/t}\,dt$.